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标准操作规程

(Standard Operating Procedure)

临床病理实验室工作流程和项目的实施

题目: Clinical Pathology Laboratory Workflow and Project
Title: Implementation

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临床病理实验室工作流程和项目的实施

1 目的

本 SOP 描述了临床病理实验室的工作流程和规范安全性评价研究临床病理项目的实施。

2 适用范围

适用于所有临床病理实验室的工作人员。

3 职责

职责	职责描述
部门负责人	确保参与试验的人员已进行合适的培训并通过考核，总体把握部门各项工作。负责协调与专题负责人在临床病理的内容沟通和项目的运行，确保项目检测数据的真实有效及分析报告的科学。
临床病理技术员	需严格按照各 SOP 要求进行查看样品状态与标识，并根据采样记录核对样本数量。标本检测、记录整理、仪器维护等操作，确保检测结果的准确和仪器状态的良好。

4 定义

无。

5 程序

5.1 项目实施程序

5.1.1 项目前期准备与检测前材料准备

(1) 临检部门负责检测的试验在试验前由部门负责人指定人员参加方案培训，针对临床病理相关检测内容和流程提出建议。

(2) 邮件收到试验方案及试验日程安排后，根据试验方案制定检测计划，并做好试验前的准备工作，如方法开发与验证、试剂与耗材的备库。

(3) 根据试验的实际情况决定是否有必要将该试验的研究背景、药物特点、检测周期、预期结果、特殊指标检测及检测完毕后标本是否需要特殊保存等情况进行说明或培训。

(4) 使用 Provantis 系统采集临检数据的专题（检疫期筛选动物、兽医诊治及 SD 要求加测等特殊方案定义的检测除外）需在 Provantis 中提交申请，技术人员登陆 Provantis 系统，进入“临床病理学”模块，选择“计划检验申请”或“非计划检验申请”，选择发送检验申请的研究，日期，动物，检验项目等，生成检验申请。在 Provantis 中提交申请时，Provantis 中方案定义必须按照方案中规定的项目进行定义，不得多选漏选检测项目，确

保申请信息准确无误。

(5) 根据检测申请信息提前准备样本标签、采集管及相关记录表。粘贴标签准备采集管时,注意核对标签样本信息(专题号、动物号、日期、样本类型)及采集管质量(密封性、有无破损、抗凝剂的量等),确保信息无误,质量合格。如有需要,可根据方案采集量要求及抗凝剂比例要求,自行配制抗凝剂制备采血管(如枸橼酸钠抗凝管),并填写“试剂配制记录表(SPAT-011-F1)”。

(6) 申请检测指标的核对:在 Provantis 系统或仪器系统申请检测指标后,需由第二人按照试验方案进行检测指标、动物信息的核对,确保无误,并填写 QC 表(SMAN-044-F1)。

5.2 标本接收与处理

5.2.1 标本接收

标本采集完成后,临床病理部应有临检技术员负责接收标本。具体参照 SMAN-033《试验材料的转移与接收》。标本接收人员应核对样本试验编号和动物号、样本数量及样本状态(标本采集量应符合方案规定要求,抗凝全血无凝块,离心后血清/血浆应为淡黄色/黄色透明/半透明液体)。核对完成后,在“临床病理标本前处理记录(SCLI-001-F1)”上做好相应记录。

如发现不合格样本(如样本管上信息与申请信息不一致;采集容器不正确,如血清样本用了抗凝管;未按 SOP 要求及时送检,导致未能在规定时间内对样本进行处理和检测;目测样本量不足或上机检测时仪器提示样本吸样量不足导致无法正常检测;目测观察到抗凝标本有凝块或上机检测时仪器提示样本有微凝集导致结果或曲线异常;离心后发现血清/血浆有较严重的溶血、黄疸或脂血,或上机检测时仪器提示相应的信息导致结果异常;等),临检技术人员应及时联系专题负责人(SD),如需重新采集标本,应由 SD 批准后通知负责该专题的试验技术员。重新采集的标本在 provantis 中或“临床病理标本前处理记录(SCLI-001-F1)”上注明重新采集。部分不合适的样本由于难以取材等一些特殊原因,经与 SD 协商仍需继续检测时,应在 provantis 系统结果备注信息栏内注明样本不合格状态。不合格样本信息需在“临床病理标本前处理记录(SCLI-001-F1)”中做好相应记录。

5.2.2 标本处理

参见 SCLI-004《临床病理标本的制备与编号》进行不同类型标本的处理。

5.3 标本检测

(1) 核对检测指标:样本检测前,由操作人员根据试验方案检查指标内容是否正确。

(2) 仪器准备:试验开始前做好仪器的开机维护,并在相关的仪器使用记录表上记录。确保在样本分析前做好仪器的准备工作。

(3) 定标及校准:生化仪、血凝仪、血球仪等定量检测项目,在试剂更换批号后或仪器维保更换关键部件后均需重新定标或校准,定标产生的数据实时打印并签字确认,随仪器使用记录进行年度归档。

(4) 质控品检测：质控品使用应严格按照其说明书要求。为保证检测质量，每天应在样本检测开始前做相应仪器的质控。如果当天相应仪器的样本检测量超过 20 例，或者仪器运行检测的时间超过 8 小时，则应在试验检测结束后加做一次关机前质控。质控品应选择两个或两个以上浓度水平（特殊项目或厂家无法提供相应浓度质控品除外）。样本检测开始前质控结果合格方可进行样本检测，关机质控结果（如有）合格方可结束当天试验。试验中操作人员如发现可疑情况（如仪器故障、试剂问题等），可视情况增加质控检测的次数。如当天样本检测前或者结束后的质控不合格，应分析失控原因后重新检测。试验中或试验后质控失控纠正后，还需对失控前检测的样本进行评估处理，同时报告实验室负责人进行审核。在 provantis 系统中对质控不合格原因以及质控纠正措施进行详细描述，无法使用 provantis 系统的仪器应在“室内质控失控记录（SCLI-001-F2）”上记录。具体按照 SCLI-002《临床病理质控血清复溶制备与室内质控规则》进行质控的分析、评价及失控处理。

(5) 上机检测：按照各类仪器的 SOP 进行操作分析。样本上机检测时，由仪器自动识别样本条码并按照识别顺序进行检测，如果仪器无法识别样本条码而需要人工录入时，仪器分析编号由临检技术员进行编写，每次测试均应将动物号和仪器分析编号的对应情况列表说明，并记录在“临床病理标本前处理记录”（SCLI-001-F1）中。进行检测。如果专题负责人对标本检测顺序有特殊要求，应在试验方案中说明。

5.4 结果的审核和复查

检测完成后检验相关人员要对结果进行审核。认真核对检测结果有无超检测线、标本状态报警、仪器状态异常报警等情况。按照 SCLI-003《临床病理标本的重复测定准则》对异常结果标本进行复检。技术人员发现异常结果时应及时向专题负责人或部门负责人报告，试验中具有典型特点的异常结果也应及时报告专题负责人。

5.5 结果的发布

(1) 不上 Provantis 系统的试验，仪器上结果审核无误后打印纸质报告，操作者及审核者对纸质报告审核签字确认后，将纸质报告作为原始数据交由专题负责人。

(2) 使用 Provantis 系统进行临床病理数据采集的，各仪器的检测结果将实时自动传输至 Provantis 系统中，当因特殊原因无法实时传输时，需有审核者对仪器中的结果进行审核，确保结果无误后再人工传输或回录至 Provantis 系统。试验人员对 Provantis 系统中的结果进行备注、数据质量标识和数据替代标识后，由审核人员审核结果并进行电子签名，确保结果无误后发布结果。结果发布后专题负责人可在 Provantis 系统试验模块进行临检数据查阅。

5.6 标本保存

血清或血浆应按照方案要求进行保存，如需要分装择期进行检测时，需要记录在“临床病理标本分装记录（SCLI-001-F3）”中，保存期限至少至该研究完成，与委托方确

定标本处理方式。其他类型的标本如果试验未进行标本保存规定的可不进行保存。标本保存和处理情况应分别被记录在“临床病理标本保存记录（SCLI-001-F4）”和“临床病理标本处理记录（SCLI-001-F5）”中；

5.7 记录的整理与一般资料的归档

(1) 检测记录完成后需在检测者处签名确认，并由核对者核对后签名。当日记录整理完毕，可由临床病理部检测技术员暂存或交给专题负责人保管。

(2) 临床病理部一般资料包括试验原始记录、试剂复溶记录、标本保存记录、标本处理记录、质控失控记录、检测仪器的使用维护保养记录、检验试剂的说明书（同一版本的说明书只用归档一次），试剂的出厂检测报告（同种试剂只归档一次），质控品及标准品的说明书及靶值、SOP 偏离记录、室间质评（能力验证）记录及不满意结果处理记录、试验中产生的其他记录资料等。临床病理部一般资料按年度进行归档，原始记录按试验归档。

6 意外事件处理

A. 仪器故障紧急处理

紧急联系仪器工程师，进行远程指导或者到场检查维修；

样本检测：检测中断后，将样本合理存放（按照 SOP “SCLI-004-V3.0 临床病理标本的制备与编号”中样本存放要求）；如仪器长时间无法恢复正常，将样本及时外送至备用合作机构实验室进行检测；

恢复检测：仪器故障维修后，需进行仪器的校准、质控，确保仪器性能状态正常后方可恢复样本检测。重大故障维修后，由工程师评估是否需要进行仪器的性能验证，验证通过后再恢复样本检测。

B. 停电、停水应急处理

评估停电停水时间：与运维部联系，确认停水停电时间，短时间停电，UPS 电源启动可为检测仪器提供 30-60 分钟的缓冲供电；实验室需常备一定量的蒸馏水/纯水，用于应急配制试剂和清洁；

保护样本与数据：停电时间超过 UPS 电源可支撑时间时，应暂停样本检测，将正在运行的仪器按 SOP 进行紧急关机，最大限度保护仪器和保存数据，记录所有中断的样本检测批次；

样本检测：检测中断后，将样本合理存放（按照 SOP “SCLI-004-V3.0 临床病理标本的制备与编号”中样本存放要求）；如仪器长时间无法恢复正常，将样本及时外送至备用合作机构实验室进行检测；

通知与沟通：立即通知实验室负责人、SD 等，评估对项目时间线的影响；

恢复检测：水电恢复后，按顺序启动设备，执行开机程序，开机维护，质控检测，确保仪器状态正常、质控合格后方可恢复样本检测。

7 相关文件

文件编号	SCLI-004	文件标题	临床病理标本的制备与编号
	SCLI-002		临床病理质控血清复溶制备与室内质控规则
	SCLI-003		临床病理标本的重复测定准则
	SMAN-033		试验材料的转移与接收
表格编号	SCLI-001-F1	表格标题	临床病理标本前处理记录
	SCLI-001-F2		室内质控失控记录
	SCLI-001-F3		临床病理标本分装记录
	SCLI-001-F4		临床病理标本保存记录
	SCLI-001-F5		临床病理标本处理记录
	SPAT-011-F1		试剂配制记录表
	SMAN-044-F1		QC 表
模板编号	无	模板标题	无

8 参考资料

- (1) 21 CFR Part58 Good Laboratory Practice for Nonclinical Laboratory Studies
- (2) OECD Good Laboratory Practice for Nonclinical Laboratory Studies
- (3) 中国国家食品药品监督管理局“药物非临床研究质量管理规范”

9 SOP修订记录

修订历史

SOP 现版本号	旧版本生效日期	修订者	修订内容描述
V2.0	2024-05-14	任晓庆	1. 5.3 标本检测中，第 4 项修改为：检测数超过 40 或者首次质控时间超过八小时则需在试验结束后加做一次质控，质控通过后方可结束操作； 2. SOP 统一双语化
V3.0	2024-10-01	任晓庆	1. 修改 5.3 中第(4)质控品检测内容。 2. 增加及修改 5.1.1 中的第(4)及(5)的内容。 3. 修改 5.5 结果的发布。 4. 在 6 相关文件中增加表格。
V4.0	2025-04-17	任晓庆	1. 5.1.1 (5) 中增加“核对样本标签及采样管信息”内容。

			2. 在 5.1.1 中增加 (6) 申请检测指标的核对。 3. 增加 6 意外事件处理。 4. 在 7 相关文件中增加 QC 表 (SMAN-044-F1)。
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Clinical Pathology Laboratory Workflow and Project Implementation

1 Purpose

This SOP describes the workflow of clinical pathology laboratories and standardizes the implementation of clinical pathology items for safety evaluation studies.

2 Scope

Applicable to all staff of clinical pathology laboratory.

3 Role

Role	Responsibility
Department Head	Ensure that the personnel participating in the trial have been appropriately trained and passed the assessment, and have a general grasp of the work of the department. Be responsible for coordinating the content communication with the study director in clinical pathology and the operation of the project to ensure the authenticity and validity of the test data of the project and the scientificity of the analysis report.
Clinical Pathology Technician	Check the sample status and identification in strict accordance with each SOP and check the number of samples according to the sampling record. Sample test, record arrangement, instrument maintenance and other operations to ensure the accuracy of test results and good instrument status.

4 Definitions

NA.

5 Procedures

5.1 Project Implementation Procedures

5.1.1 Pre-Project Preparation and Pre-Test Material Preparation

- (1) The personnel designated by the department head for the testing performed by clinical testing department shall participate in protocol training before the trial, and put forward suggestions on the content and process of clinical pathology-related testing.
- (2) After receiving the study protocol and study schedule by mail, develop the test plan according to the study protocol, and make preparations before the trial, such as method development and verification, and storage of reagents and consumables.
- (3) According to the actual situation of the study, decide whether it is necessary to explain or train the research background, drug characteristics, detection cycle, expected results, detection of special indicators of the trial and whether the samples need special preservation

after the completion of detection.

(4) For the study using Provantis system to collect the clinical test data (except for screening animals during quarantine period, veterinary diagnosis and treatment and additional testing required by SD and other tests defined in special protocol), an application needs to be submitted in Provantis. The technician logs in to Provantis system, enters "clinical pathology" module, selects "application for scheduled test" or "application for unscheduled test", selects the study with test application, date, animal, item, etc., to generate the test application. When submitting an application in Provantis, the protocol definition in Provantis must be performed in accordance with the items specified in the protocol, and do not over-choose or omit the items to ensure that the application information is correct.

(5) Prepare sample labels, collection tubes, and related record forms in advance based on the testing application information. When preparing the collection tube by pasting the label, pay attention to checking the sample information on the label (topic number, animal number, date, sample type) and the quality of the collection tube (sealing, whether there is damage, amount of anticoagulant, etc.) to ensure that the information is correct and the quality is qualified. If necessary, according to the collection volume requirements and anticoagulant ratio requirements of the plan, prepare anticoagulant tubes (such as sodium citrate anticoagulant tubes) and fill out the "Reagent Preparation Record Form (SPAT-011-F1)".

(6) Verification of applied testing indicators: After applying for testing indicators in the Provantis system, a second person needs to verify the testing indicators and animal information according to the experimental plan to ensure accuracy, and fill out the QC form (SMAN-044-F1).

5.2 Receipt and Handling of Specimens

5.2.1 Specimen Receipt

Upon completion of specimen collection, the Clinical Pathology Laboratory Technician is responsible for receiving the specimen. Refer to SMAN-033 "Study Materials Transport And Reception Record" for details. The sample receiving personnel shall check the sample test number and animal number, sample size and sample status (the sample collection volume shall meet the requirements specified in the protocol, there is no clot in anticoagulated whole blood, and the serum/plasma shall be light yellow/yellow transparent/translucent liquid after centrifugation). Following completion of reconciliation, appropriate documentation will be documented on the "Pre treatment record of clinical test samples (SCLI-001-F1)".

If unqualified samples are found (for example, the information on the sample tube is inconsistent with that on the application information; the collection container is incorrect, for example, the anticoagulant tube is used for serum samples; the samples are not submitted in time according to SOP requirements, resulting in failure to process and test the samples within the specified time; the sample size is insufficient by visual inspection or the sample suction volume is insufficient by the instrument during on-board testing, resulting in failure to detect the samples normally; the anticoagulant sample is observed by visual inspection to have clots or the sample is indicated by the instrument during on-board testing, resulting in abnormal

results or curves; the serum/plasma is found to have severe hemolysis, jaundice or lipemia after centrifugation, or the corresponding information is indicated by the instrument during on-board testing, resulting in abnormal results; etc.), the clinical technicians shall timely contact the study director (SD). If re-collection is required, the SD shall notify the test technician responsible for the study after approval. Recollected specimens were noted as recollected in Provantis or on the "Clinical Pathology Specimen Pretreatment Record (SCLI-001-F1)". Where some unsuitable samples still need to be tested after consultation with SD due to some special reasons such as difficulty in obtaining samples, the unqualified status of samples shall be indicated in the Results Remarks Information Column of Provantis System. Rejected sample information is documented on the "Pre treatment record of clinical test samples" (SCLI-001-F1).

5.2.2 Specimen Processing

Refer to SCLI-004 "Preparation and numbering of clinical pathological specimens" for the processing of different types of specimens

5.3 Specimen Testing

- (1) Check the test indicators: before sample test, the operator checks whether the indicators are correct according to the study protocol.
- (2) Instrument preparation: Perform startup maintenance of the instrument before the start of the test and record it on the relevant instrument use record form. Ensure that the instrument is prepared before sample analysis.
- (3) Calibration and Adjusting: For quantitative test items such as biochemical analyzer, coagulometer and hemocytometer, it is required to re-calibrate or calibrate after the reagent changes the batch number or after the instrument maintenance replaces the key components. The data generated by calibration shall be printed and signed for confirmation in real time, and archived annually with the instrument use records.
- (4) Control detection: Control shall be used in strict accordance with the requirements in the instructions for use. In order to ensure the quality of testing, quality control of corresponding instruments should be performed every day before the start of sample testing. If there are more than 20 samples detected by the corresponding instrument on that day, or the instrument has been running for more than 8 hours, an additional quality control of shutdown should be conducted after the end of the experimental testing. Two or more concentration levels should be selected for quality controls (except for special items or quality controls whose corresponding concentration cannot be provided by the manufacturer). Samples can be detected only when the quality control results are qualified before the start of sample detection, and the shutdown quality control result (if any) must be qualified before the test of the day can be ended. When the doubtful circumstances (such as instrument malfunctions, reagent problems, etc.) are noted, additional quality control test shall be performed. If the quality control before or after the sample test on that day fails, the cause of loss of control shall be analyzed and the test shall be repeated. After correcting the quality control loss during or after the experiment, it is necessary to evaluate and process the samples tested before the

loss of control, and report to the laboratory manager for review. Reasons for quality control failure and corrective actions for quality control need to be described in detail in the Provantis system, which also should be documented on the "Record of Loss of Control of Internal Quality Control (SCLI-001-F2)" if instruments unable to use the Provantis system. According to SCLI-002 "Clinical Pathology Quality Control Serum Reconstitution Preparation and Internal Quality Control Rules", quality control analysis, evaluation and out-of-control handling should be performed.

(5) Testing: Perform testing and analysis according to the SOP of each instrument.

When samples are detected on the machine, the sample barcode is automatically identified by the instrument and detected according to the identification order. If the instrument cannot identify the sample barcode and manual entry is required, the instrument analysis number is prepared by the clinical technician. For each test, the corresponding situation list of animal number and instrument analysis number should be described and recorded in the "Record of Pretreatment of Clinical Pathological Specimens" (SCLI-001-F1). Performing the test. If the Study Director has specific requirements for the order of specimen testing, this should be stated in the protocol.

5.4 Review of Results

The inspection related personnel shall review the results after the completion of detection. Carefully check whether the test results have such conditions as exceeding the test line, sample status alarm and abnormal instrument status alarm. Specimens with abnormal results were retested according to SCLI-003 Criteria for Repeated Determination of Clinical Pathology Specimens. Technicians shall timely report abnormal results to the study director or department head when discovering them, and timely report abnormal results with typical characteristics in the test to the study director.

5.5 Release of Results

- (1) For trials not involving Provantis system, a paper report will be printed after verifying the results on the instrument. After the operator and reviewer have signed and confirmed the paper report, it will be submitted as the original data to the study director.
- (2) For the collection of clinical pathological data using Provantis system, the detection results of each instrument will be automatically transmitted in real-time to the Provantis system. When real-time transmission is not possible due to special reasons, a reviewer needs to review the results in the instrument to ensure that they are correct before manually transmitting or recording them back to the Provantis system. After the test personnel comment on the results, data quality identification and data substitution identification, the reviewer reviews the results and electronically signs them to ensure that the results are correct and then publishes the results. After the results are released, the study director can consult the clinical test data in the Provantis system test module.

5.6 Sample Storage

Serum or plasma should be stored according to the protocol. If dispensing is required for

elective testing, it should be recorded in the “Clinical Pathological Specimen Sub-packing Record (SCLI-001-F3)”. The storage period should be at least until the study is completed to determine the specimen processing with the sponsor. Other types of specimens may not be preserved if specimen preservation is not specified in the study. Specimen storage and processing shall be documented on the “Sample Keeping Records of Clinical pathology (SCLI-001-F4)” and “Sample Handling Records of Clinical pathology (SCLI-001-F5)”, respectively.

5.7 Records Collation and Archiving of General Information

(1) After the completion of the test record, the tester shall sign and confirm it, and check it by the checker and sign it. On that day, the records are sorted out and temporarily stored by the testing technician of Clinical Pathology Department or delivered to the study director for storage.

(2) General data of clinical pathology department include original study records, reagent reconstitution records, specimen preservation records, specimen processing records, quality control out-of-control records, use and maintenance records of detection instruments, instructions of test reagents (the instructions of the same version are only archived once), delivery test reports of reagents (the same kind of reagents are only archived once), instructions and target values of controls and standards, SOP deviation records, inter-laboratory quality assessment (capacity verification) records and unsatisfactory results processing records, and other records generated in the study. General data of clinical pathology department were archived annually, and original records were archived as per the study.

6 Handling of Unexpected Events

A. Emergency handling of instrument malfunctions

Urgently contact the instrument engineer for remote guidance or on-site inspection and maintenance;

Sample testing: After interruption of testing, store the samples properly (according to SOP "SCLI-004-V3.0 Clinical Pathology Standards Sample storage requirements in the preparation and numbering of this document; If the instrument cannot be restored to normal for a long time, the sample should be promptly sent to a backup cooperative laboratory for testing;

Recovery testing: After repairing the instrument malfunction, it is necessary to calibrate and quality control the instrument to ensure that its performance is normal before resuming sample testing. After major fault repairs, engineers evaluate whether instrument performance validation is necessary, and only resume sample testing after validation is passed.

B. Emergency response to power and water outages

Assess the duration of power and water outages: Contact the operations department to confirm the duration of power outages, short-term power outages, and UPS power supply

Startup can provide 30-60 minutes of buffer power supply for the detection instrument; The laboratory needs to keep a certain amount of distilled water/pure water on hand for emergency preparation of reagents and cleaning;

Protecting samples and data: When the power outage exceeds the support time of the UPS

power supply, sample testing should be paused, and the running instruments should be urgently shut down according to SOP to maximize instrument protection and data preservation. All interrupted sample testing batches should be recorded;

Sample testing: After the testing is interrupted, store the samples properly (according to the sample storage requirements in SOP "SCLI-004-V3.0 Preparation and Numbering of Clinical Pathological Specimens"); If the instrument cannot be restored to normal for a long time, the sample should be promptly sent to a backup cooperative laboratory for testing;

Notification and Communication: Immediately notify the laboratory manager, SD, etc. to assess the impact on the project timeline;

Recovery testing: After the water and electricity are restored, start the equipment in sequence, execute the startup program, perform startup maintenance, and perform quality control testing to ensure that the instrument is in normal condition and the quality control is qualified before resuming sample testing.

7 Relevant Documents

Document Number	SCLI-004	Document Title	Preparation and numbering of clinical pathological specimens
	SCLI-002		Clinical pathology quality control serum reconstitution preparation and internal quality control rules
	SCLI-003		Criteria for Repeated Testing of Clinical Pathology Specimens
	SMAN-033		Transfer and Receipt of Test Materials
Form Number	SCLI-001-F1	Form Title	Retreatment record of clinical pathological specimens
	SCLI-001-F2		Indoor QC runaway record
	SCLI-001-F3		Clinical Pathology Sample Dispensing Record
	SCLI-001-F4		Clinical Pathology Specimen Retention Record
	SCLI-001-F5		Clinical Pathology Specimen Processing Record
	SPAT-011-F1		Reagent Preparation Record Form
	SMAN-044-F1		Quality Control Form
Template Number	NA	Template Title	NA

8 References

- (1) The United States Food and Drug Administration (US FDA) Good Laboratory Practice for Nonclinical Laboratory Studies (21 CFR Part 58).
- (2) OECD Good Laboratory Practice for Nonclinical Laboratory Studies.
- (3) National Medical Products Administration (NMPA) Good Laboratory Practice for Nonclinical Laboratory Studies.

9 SOP Revision History

Revision History

SOP current version No.	Effective date of the old version	Reviser	Description of revision
V2.0	2024-05-14	Xiaoqing Ren	1. 5.3 In specimen testing, item 4 is modified to read: if the test number exceeds 40 or the first quality control time exceeds 8 hours, a quality control should be added after the test, and the operation can be finished after the quality control passes; 2. SOP unified bilingualism.
V3.0	2024-10-01	Xiaoqing Ren	1. Modified the content of quality control testing in section (4) of 5.3. 2. Added and modified the content of (4) and (5) in section 5.1.1. 3. Modified 5.5 release of results. 4. Added form in 6 Relevant Documents.
V4.0	2025-04-17	Xiaoqing Ren	1. Added the content of "verifying sample labels and sampling tube information" in section 1.5.1.1 (5). 2. Added (6) Verification of applied testing indicators in section 5.1.1. 3. Added 6 Handling of Unexpected Events. 4. Added Quality Control Form (SMAN-044-F1) in 7 Relevant Documents.